Scaling the Attack Breaks Defense Containment in Task-Level Radar–Jammer Self-Play

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***Abstract*—Modern multifunction phased-array radars must schedule heterogeneous missions while facing coordinated, adaptive interference. Link-level signal-to-interference ratios do not by themselves reveal whether a radar network completes its deadline-constrained mission load. We introduce FluxPhased, a physics-grounded task-level benchmark for adversarial co-adaptation between multifunction radars and cooperative jammers. Missions carry service and bearing identities, deadlines, and exact detection credit; the link model combines generalized planar-array responses with finite jammer energy budgets. A pre-training contestability test and a four-view evaluation protocol separate natural scheduling loss from adversarial loss.**

In a one-jammer versus two-radar game, the valid 12-dB S6 runs neutralize $63.7\% \pm 1.0\%$ of the jammer's floor-adjusted disruptive power across two training seeds. When the attacker team is doubled under the same 63-token team budget, three converged seeds yield a stable 2v2 equilibrium with head-to-head mission drop 0.3366 ± 0.0143 and neutralization $23.0\% \pm 1.1\%$. Radar competence against idle jammers remains low, showing that the degradation is an offensive scaling effect rather than a corresponding defender collapse. A co-located-jammer control attributes most of the change to attacker count under the S7 separated-radar geometry (neutralization 28.4%), with cross-fire geometry contributing a further reduction to 24.2%. Finally, a lightweight opponent-class mixture reduces singleton relative leverage from 0.502 to 0.262 at 75% singleton exposure, with the largest neutralization value at 50% exposure. The results establish a measurable separation between equilibrium stability, attack scaling, and opponent-distribution robustness.

I. Introduction

Electronic-support and electronic-attack systems interact through a resource-constrained loop: an interferer chooses where and when to spend energy, while a multifunction radar chooses which service and bearing to observe before each mission deadline. A high instantaneous JNR can be operationally irrelevant if the radar detects the corresponding target in another dwell, whereas a modest interference pulse can matter when it arrives at the only vulnerable point in a deadline window. The operational outcome is therefore mission completion, not a single link-budget scalar.

This distinction exposes a gap between two established lines of work. Radar–jammer learning studies formulate waveform, frequency, or power decisions as sequential games. Multifunction-radar scheduling studies optimize scan and track allocation under time or power constraints. These lines of work leave open a system-level question: how does the balance of an adaptive game change when the number of coordinated attackers increases while the defender's mission workload and total interference budget remain fixed?

We answer this question with a controlled S6-to-S7 progression in FluxPhased. S6 contains one learned jammer and two learned radar heads. S7 contains two parameter-shared jammer agents and two parameter-shared radar agents. The S7 team receives a 63-cell activation budget, split 32/31 between jammers. This controls total jammer activation while the radar-site geometry changes from S6 to S7; the matched co-located-jammer control isolates the additional cross-fire component within the S7 geometry.

The main result is a change in containment, not a corresponding failure of learning. Across three S7 seeds trained to 3000 iterations, head-to-head mission drop is 0.3366 ± 0.0143, compared with 0.0888 ± 0.0053 in the valid two-seed 12-dB S6 baseline. The floor-adjusted fraction of marginal jammer disruption removed by learned radars falls from $63.7\% \pm 1.0\%$ to $23.0\% \pm 1.1\%$. The radar floor against idle jammers remains low (0.0194 ± 0.0008 in S7 versus 0.0275 ± 0.0110 in valid S6), and the S7 trajectory is stationary from approximately iteration 1700 through 3000. Thus, the second attacker changes the stable equilibrium's scale while preserving low idle-jammer loss and training stability.

This paper makes four contributions:

1) We provide a task-level, bearing-aware benchmark in which deadline-constrained mission loss is coupled to array-factor link physics, finite energy, and two-team MAPPO.
2) We quantify a benchmark-level attacker-count scaling effect: with total jammer activation controlled, doubling attackers reduces defense containment by roughly two-thirds, with replication across three converged S7 seeds and a two-seed valid S6 reference.
3) We use a matched co-located-jammer control to decompose the mechanism under the separated S7 radar geometry. Two jammers already reduce containment to 28.4%; cross-fire adds a secondary reduction to 24.2%.
4) We test a minimal opponent-class mixture. The singleton-to-pair disruption ratio falls from 0.502 to 0.262 as singleton exposure increases, identifying opponent-class mixing as a practical robustness intervention while documenting its training-budget confound.

The rest of the paper defines the benchmark and evaluation metric, presents convergence and scaling results, tests the count–geometry decomposition, analyzes opponent-specific adaptation, and evaluates the lightweight mixing intervention. We use "drop" for the fraction of eligible missions that are not completed before their deadlines; higher drop therefore indicates greater jammer-side success.

II. Related Work

A. Learning in radar–jammer games

Recent radar anti-jamming studies have used multi-agent reinforcement learning and game-theoretic formulations to adapt waveform, frequency, and power decisions. Neural fictitious self-play (NFSP) has been applied to imperfect-information games, and related radar studies use game-based learning for frequency-agile and waveform decisions. These studies establish adaptive radar–jammer interaction as a useful game formulation, but their principal outcomes are link-level quantities such as SINR, detection probability, or waveform utility. FluxPhased complements this line with a mission-level endpoint: a target is successful only when the appropriate service detects the appropriate bearing within a deadline window [1].

B. Multifunction-radar resource management

A separate literature treats multifunction-radar scheduling as an MDP or constrained MDP. Deep Q-learning, actor–critic methods, PPO, SAC, and model-based planning have been applied to scan–track allocation, congested task scheduling, and time or power budgeting. These approaches address the central scheduling trade-off that a radar cannot attend to every task at every instant. They generally use fixed or scripted interference conditions, however, and therefore do not measure how a learning radar changes when the adversary co-adapts. Our benchmark retains the scheduling structure of this literature while adding a learning jammer team and an explicit natural-floor control.

C. Self-play, specialization, and leagues

Self-play can produce strong coordination but can also specialize agents to the distribution of partners or opponents encountered during training [1]–[3]. Role-oriented multi-agent learning relates team performance to emergent division of labor, and population-based training addresses opponent diversity rather than a single opponent trajectory. The hierarchical macro-strategy model for MOBA games likewise separates high-level target allocation from low-level execution and uses cross-agent coordination [4]. We borrow these ideas as motivation; R5 tests a minimal opponent-class mixture, not a full population-based league.

D. Position of this work

FluxPhased lies at the intersection of these communities. Its agents learn link-facing actions, but the reported endpoint is task completion under service, bearing, deadline, and energy constraints. This enables a controlled question that link-level or single-sided scheduling studies do not answer: whether an apparent defensive advantage survives attacker-count scaling. The resulting comparison is designed around three controls: the idle-radar floor, a scripted-radar jammer view, and a co-located-jammer ablation. The controls let us separate absolute loss, adversarial marginal loss, equilibrium stability, and robustness to a changed opponent class.

III. Benchmark and Method

A. Task-level mission model

At each time step, each service can receive an arrival. An admitted mission is represented by

$$m = (s, a, t_{\mathrm{arr}}, t_{\mathrm{dl}}, n_{\mathrm{det}}), \tag{1}$$

where s is the service, $a \in \{0, \ldots, 4\}$ is its azimuth sector, and the deadline is at most six steps after arrival. A mission succeeds when the radar team obtains one detection with the exact (s, a) identity before the deadline. The tracker credits the oldest incomplete mission with the matching service and bearing; detections at the deadline are processed before finalization. The mission drop ratio is

$$d = \frac{N_{\mathrm{timeout}} + N_{\mathrm{admission\ reject}} + N_{\mathrm{horizon\ failure}}}{N_{\mathrm{eligible}}}. \tag{2}$$

A larger value means that the jammer team denied more missions. We retain completed-but-not-yet-finalized missions in the queue, which makes the deadline accounting explicit and auditable.

B. Bearing-aware link physics

Each agent selects a beam from a 5×5 azimuth–elevation grid. A jammer selects a Bernoulli mask over 25 array cells and a categorical beam. For a steering direction (u_b, v_b) and target (u_t, v_t), the normalized planar-array response is separable:

$$G_{\mathrm{AF}}(b,t) = 10\log_{10}\left(\frac{A_5(u_t - u_b)^2}{5^2}\right) + 10\log_{10}\left(\frac{A_5(v_t - v_b)^2}{5^2}\right), \tag{3}$$

where A_5 is the finite-array response evaluated with the same float32 implementation for all experiments. For jammer k and radar r, the received JNR includes jammer cell aggregation, transmit and receive array gains at their pair bearing, service-dependent path loss and receiver gain, and polarization loss. Independent jammers are combined as incoherent power:

$$\mathrm{JNR}_r = 10\log_{10}\sum_k 10^{\mathrm{JNR}_{kr}/10}. \tag{4}$$

The effective SNR is

$$\mathrm{SNR}_{\mathrm{eff},r} = 10\log_{10}\left(\frac{10^{\mathrm{SNR}_0/10}}{1 + 10^{\mathrm{JNR}_r/10}}\right) + G_{\mathrm{coh}}, \tag{5}$$

with the idle case defined by $\mathrm{JNR}_r = -\infty$. A mission at azimuth a is detected with

$$p_r(s,a) = \sigma\left(\frac{\mathrm{SNR}_{\mathrm{eff},r} + G_{\mathrm{target}}(b_r, a) - \tau}{w}\right), \tag{6}$$

where $\tau = 15$ dB and $w = 3$ dB. This couples radar pointing to both useful detection gain and exposure to interference.

C. S6 and S7 games

S6 uses one jammer against two co-located learned radar heads. S7 uses two parameter-shared jammers at azimuths $+60^\circ$ and -60° and two radar heads at $+20^\circ$ and -20°. Both stages use the validated $\mathrm{SNR}_0 = 12$ dB regime and $P_{\mathrm{jam}} = 0.1$ W per active cell. The S7 team receives a total 63-token budget, split 32/31 between jammers; the S6-to-S7 comparison therefore changes attacker count and radar-site geometry while holding total S7 jammer activation fixed.

Each side uses a parameter-shared multi-head actor. Radar agents choose beam and service; jammer agents choose cell mask and beam. A centralized privileged critic receives the concatenated public observations of its team during training, while the deployable actor receives only its local observation and last-action ESM channels. PPO uses generalized advantage estimation, clipped likelihood ratios, per-head entropy schedules, and separate actor, local critic, and privileged-critic optimizers [5], [6]. Rollouts flatten agent slots in agent-major order so the shared team reward is duplicated without mixing trajectories.

D. Evaluation and containment

We evaluate four views for S7: head-to-head (h2h), learned jammers against scripted sweep radars (jvs), learned radars against idle jammers (rad-idle), and one learned jammer against the pair-trained radars (j1-only). The S6 baseline defines h2h, jvs, rad-idle, and the natural floor; it does not contain the S7-specific j1-only control. The scripted sweep against idle jammers supplies the natural floor d_{floor}. Let d_{h2h}, d_{jvs}, and d_{idle} denote the corresponding drop ratios.

$$\eta = 1 - \frac{d_{\mathrm{h2h}} - d_{\mathrm{idle}}}{d_{\mathrm{jvs}} - d_{\mathrm{floor}}}. \tag{7}$$

This metric measures the fraction of the jammer's marginal disruption that learning radars remove; it avoids crediting the radar for losses caused by its own scheduling floor. Final evaluations use 64 fixed validation scenarios and three independent action seeds (4242, 777, and 31337).

E. Pre-training contestability gate

Before training, we enumerate beam responses for each mission azimuth. The gate requires at least three of five azimuths to have best-response detection probability in $(0.05, 0.95)$, a minimum above 0.02, and a maximum above 0.8. The S7 cross-fire profile is $(0.837, 0.912, 0.989, 0.993, 0.855)$; all five sectors remain reachable and three remain contestable. The co-located-jammer control profile is $(0.921, 0.823, 0.989, 0.996, 0.991)$, which predicts a more defense-favorable game before any policy update.

IV. Experimental Protocol

A. Training and checkpoints

We use 64 fixed training scenarios and a separate 64-scenario validation manifest. Each environment rollout has 16 parallel environments and a 64-step horizon. The S7 main experiments use three random seeds (20260801, 20260802, and 20260803). Each seed trains for 1000 iterations with the normal per-head entropy schedule and continues to 3000 iterations with entropy coefficients frozen at their minimum. The continuation isolates additional optimization from a second exploration schedule. The three converged checkpoints are stored in `s7_continue2_output_seed20260801`, `s7_seed02_cont_output_seed20260802`, and `s7_seed03_cont_output_seed20260803`.

The primary S6 comparison uses the two valid 12-dB training seeds 20260730 and 20260731. An earlier S6 seed, 20260729, was trained accidentally with a 22-dB override; we exclude it from the primary comparison and retain it only as a documented regime check. The valid two-seed S6 aggregate is used wherever an S6 uncertainty is reported.

B. Mechanism ablation

The co-located-jammer control uses seed 20260811, the same 1000+1000 two-stage protocol, the same 63-token team budget, radar sites at $+20^\circ$ and -20°, and both jammer sites at $+60^\circ$. Its corrected evaluation passes the explicit `+60,+60` geometry to the evaluator. The first evaluation generated with the default cross-fire geometry is retained as an audit file and is not used in any reported result.

C. Opponent-class mixture

R5-lite trains one seed for each singleton exposure fraction $q \in \{0.25, 0.50, 0.75\}$, for 2000 iterations under the same two-stage protocol. On a fraction q of iterations, jammer 1 is forced idle and the radar team updates against the singleton opponent; the jammer-team update is skipped on those iterations. This is a minimal opponent-class mixture rather than a full population league. The reference $q = 0$ is the S7 seed-01 2000-iteration checkpoint. Because jammer updates are skipped on singleton iterations, the absolute jammer strength changes with q. We therefore treat $j1/jvs$, $h2h/jvs$, and η as the primary cross-condition readouts and present absolute drops as descriptive context.

D. Statistical reporting

For S7, the final evaluation runs four views over all 64 validation scenarios with one repetition for each of three action seeds. The S6 implementation supplies h2h, jvs, rad-idle, and the natural floor but does not supply the S7-specific j1-only control. We report the mean and standard deviation across action seeds for each training seed. Aggregate S7 values report the mean and standard deviation across the three training seeds. The natural floor is computed using scripted radar sweeps against idle jammers under the same environment regime. We retain all raw JSONL logs and final evaluation JSON files in the repository artifact tree.

E. Implementation checks

The S7 environment and physics tests contain 18 contract and physics gates. They cover action shapes, energy accounting, exact service–bearing credit, generalized array-factor consistency, two-source JNR summation, detection monotonicity,

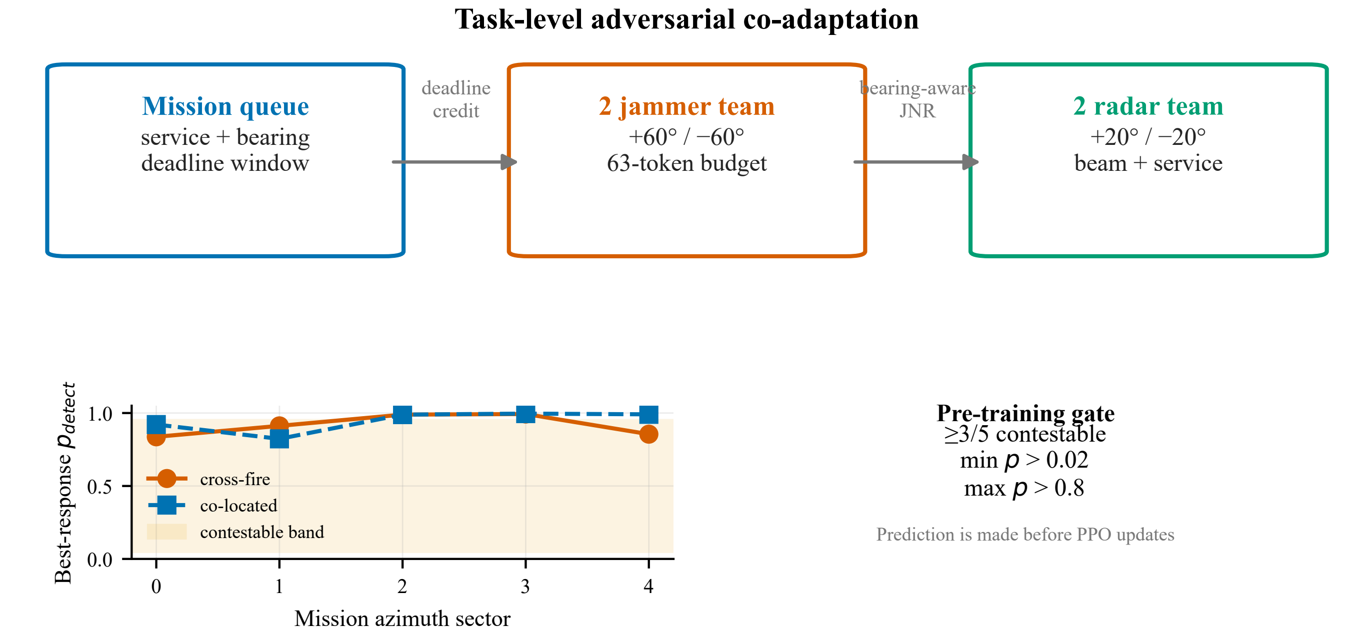

Fig. 1. Benchmark anatomy and pre-training contestability. FluxPhased couples a deadline-constrained mission queue to bearing-aware jammer–radar link physics. The lower panel reports the best-response detection profiles used to gate the cross-fire and co-located configurations before training.

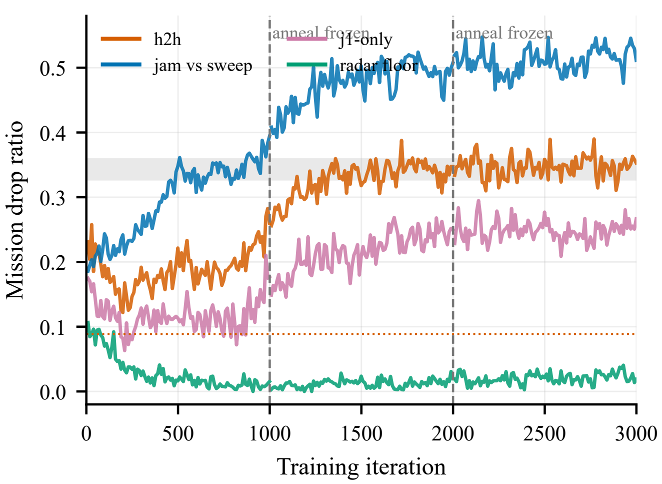

Fig. 2. Three-stage S7 trajectory for seed 20260801. Dashed lines mark the 1000- and 2000-iteration continuation boundaries; the final region is stationary across the four evaluation views.

observation channels, ledger identity, and pre-training contestability. Atomic checkpoints and a max-iteration resume criterion protect long runs from interrupted sessions. These engineering checks are separate from the scientific result and are reported to make the training trajectory reproducible.

V. Results

A. Self-play converges to stable task-level equilibria

The first question is whether mission-level self-play produces a reproducible endpoint rather than a transient training phase. Figure 2 shows the full three-stage trajectory for S7 seed 20260801. After the entropy schedule was frozen, the five 200-iteration windows from 2000 to 3000 had h2h means 0.3465, 0.3428, 0.3484, 0.3507, and 0.3463. The corresponding jam-vs-sweep means were 0.5121, 0.4956, 0.5119, 0.5107, and 0.5215. This continuation therefore rules out both an early stopping artifact and a persistent drift. The last 40 validation points at 3000 give h2h 0.3485 ± 0.0151 and jam-vs-sweep 0.5161 ± 0.0174.

The same endpoint appears across three converged S7 seeds. Full-protocol evaluation gives h2h drops of 0.3390, 0.3212, and 0.3496, with an across-seed mean of 0.3366 ± 0.0143. The idle-radar drop is 0.0194 ± 0.0008 in S7, compared with 0.0275 ± 0.0110 across the two valid S6 seeds; both values remain low. The convergence result is descriptive: it establishes a stable attractor for this benchmark and training protocol, not a proof of global Nash equilibrium.

B. Doubling the attacker reduces defense containment by two-thirds

The main comparison is summarized in Table I and Figure 3. In S6, the valid two-seed 12-dB h2h drop is 0.0888 ± 0.0053, jam-vs-sweep is 0.2751 ± 0.0110, and floor-adjusted neutralization is $63.7\% \pm 1.0\%$. In converged S7, these values become 0.3366 ± 0.0143, 0.5294 ± 0.0215, and $23.0\% \pm 1.1\%$, respectively.

The neutralization change is not a consequence of a corresponding radar collapse. Learned radars facing idle jammers retain a low drop of 0.0194 ± 0.0008 in S7, compared with 0.0275 ± 0.0110 across the two valid S6 seeds. Instead, two jammers increase the marginal disruption that survives radar adaptation. The jammer team's raw disruption against scripted sweeps rises by 92% relative to S6, while the fraction of that disruption removed by learned radars falls by approximately two-thirds. Because the comparison controls total jammer activation but also changes radar-site geometry, this is a

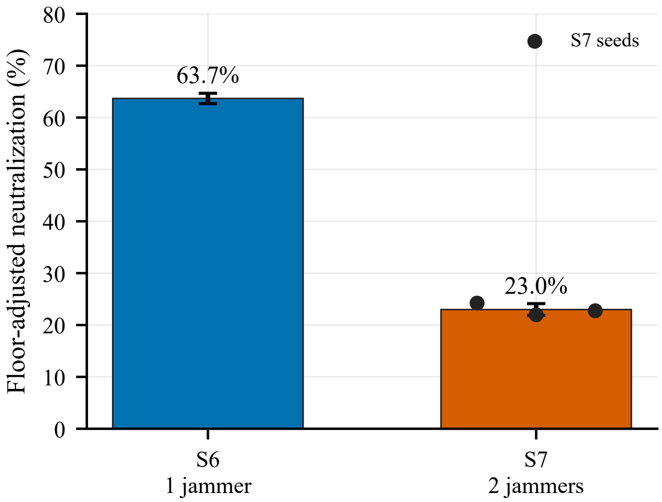

Fig. 3. Floor-adjusted neutralization for the S6 and S7 games. The dots show the three converged S7 training seeds; error bars are standard deviations across training seeds (S6 uses the two valid 12-dB seeds).

TABLE I
CONVERGED S6–S7 COMPARISON. DROP IS THE MISSION-LOSS RATIO; η IS FLOOR-ADJUSTED DEFENSE NEUTRALIZATION.

Metric	S6: 1v2	S7: 2v2
h2h drop	0.0888 ± 0.0053	0.3366 ± 0.0143
jam-vs-sweep drop	0.2751 ± 0.0110	0.5294 ± 0.0215
rad-vs-idle drop	0.0275 ± 0.0110	0.0194 ± 0.0008
η	$63.7\% \pm 1.0\%$	$23.0\% \pm 1.1\%$

benchmark-level scaling result rather than a fully isolated attacker-count intervention.

C. Attacker count is primary; cross-fire geometry is secondary

The co-located-jammer control tests whether the S7 result requires two spatially separated jammers. It places both jammer sites at $+60°$ while retaining the two radar sites at $\pm 20°$, the same physics, budget, two-stage training, and final evaluation. The corrected control yields h2h 0.2927 ± 0.0119, jam-vs-sweep 0.4947 ± 0.0026, and neutralization 28.4%.

Thus, the matched S7 geometry control shows that two jammers account for most of the containment loss within the separated-radar setting: containment is 28.4% for co-located jammers and 24.2% for cross-fire. The S6-to-S7 comparison additionally changes radar-site geometry, so the full count effect is a benchmark-level attribution rather than a perfectly isolated causal contrast. The cross-fire comparison itself is controlled for radar placement, budget, training protocol, and evaluation. The pre-specified contestability profiles and the matched control are shown in Figure 1; the converged mechanism decomposition is shown in Figure 4.

D. Pair-trained defenses trade singleton robustness for pairwise defense

The fourth view evaluates one learned jammer against radars trained in the 2v2 game. Across the three converged S7 seeds, the singleton drop is 0.2632, 0.2387, and 0.1238, with mean 0.2086 ± 0.0745. This variation is larger than the h2h variation, but all three seeds expose the same opponent-class boundary: a defense optimized against a pair is not uniformly robust to a singleton.

The seed-01 continuation makes the trade-off visible over time. As the pair-trained equilibrium develops, j1-only rises from approximately 0.12 near the 1000-iteration checkpoint to approximately 0.24 at 2000 and remains near 0.25 at 3000, while h2h settles near 0.34. Radar competence against idle jammers remains low and stable. The result is therefore an adaptation trade-off, not evidence that the radar actor or critic stopped learning.

The learned behavior is interpretable and is summarized in Figure 5.

TABLE II
R5-LITE OPPONENT-CLASS MIXTURE. ABSOLUTE DROPS ARE DESCRIPTIVE BECAUSE SINGLETON ITERATIONS SKIP THE JAMMER UPDATE; RATIO METRICS ARE THE PRIMARY COMPARISON.

q	h2h	jvs	j1-only	j1/jvs	η
0	.3282	.5043	.2532	.502	20.2%
.25	.2470	.4015	.1767	.440	23.5%
.50	.1962	.3580	.0992	.277	26.1%
.75	.1939	.3476	.0911	.262	24.6%

E. Opponent-class mixing reduces singleton leverage

The full dose-response is shown in Figure 6 and Table II.

Singleton relative leverage $d_{\mathrm{j1}}/d_{\mathrm{jvs}}$ falls monotonically from 0.502 to 0.262 and saturates between 50% and 75% exposure. The containment ratio η increases from 20.2% to 26.1% at 50% exposure before leveling at 24.6%. The h2h-to-jvs ratio is 0.651 at $q = 0$, falls to 0.548 at $q = 0.50$, and is 0.558 at $q = 0.75$. It therefore improves through the 50% condition and changes little thereafter; the ratio-level data do not show a pairwise-versus-singleton trade-off in this intervention.

R5-lite has an important design boundary. On singleton iterations, jammer 1 is forced idle and its update is skipped; consequently the jammer team receives fewer gradient updates as q increases, and absolute jvs drops from 0.5043 to 0.3476. The dose-response does not identify a causal improvement in absolute radar performance. It does support a narrower intervention claim: under this minimal mixture and fixed training budget, exposing the radar team to a singleton opponent reduces the singleton's relative leverage, with diminishing returns after 50% exposure. A gradient-aligned multi-seed experiment is needed before calling the result universal.

VI. DISCUSSION

The experiments reveal a distinction that is hidden by link-level reporting. Increasing the attacker team from one to two does not make self-play unstable; it changes how much of the jammer's marginal disruption the learned radar team can remove. Under the tested task workload, the same radar competence floor coexists with a much larger adversarial loss.

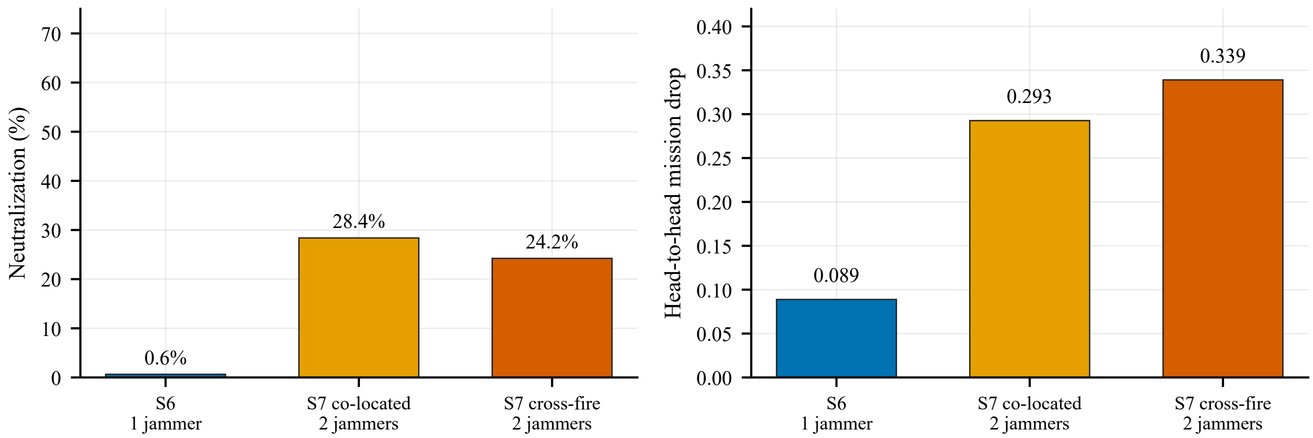

Fig. 4. Mechanism decomposition under matched task, budget, and training protocols. Two co-located jammers reproduce most of the containment loss, while cross-fire geometry adds a secondary penalty.

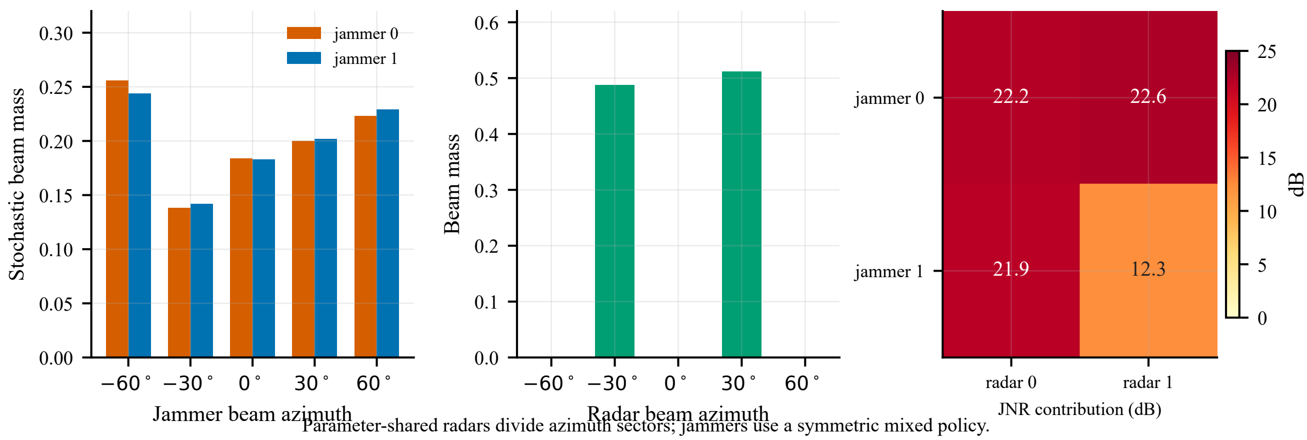

Fig. 5. Equilibrium behavior at the converged S7 checkpoint. Radar heads divide the mission plane between two azimuth sectors, while the jammer team uses a symmetric mixed beam policy. The heat map reports active-only pairwise JNR contributions in dB.

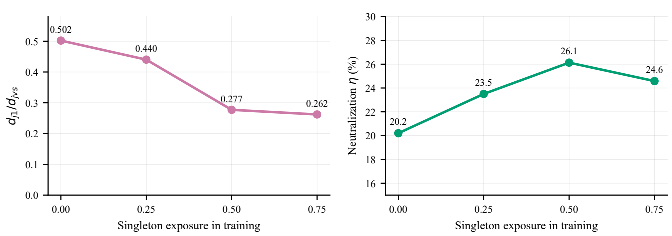

Fig. 6. R5-lite opponent-class mixing. Singleton relative leverage d_{j1}/d_{jvs} falls with singleton exposure, while floor-adjusted neutralization peaks near 50%. Absolute drops are confounded by skipped jammer updates.

The relevant design variable is therefore not only the quality of a single radar policy, but also the number and spatial arrangement of simultaneous interference sources.

The mechanism control refines this statement. Two jammers at one bearing already reduce containment from 63.7% to 28.4%. Cross-fire reduces it further to 24.2%. The benchmark therefore assigns primary responsibility to attacker count and a secondary penalty to geometry. This decomposition is more useful than attributing the entire effect to a visually appealing cross-fire explanation: a defense that survives one jammer should be stress-tested against the energy and timing consequences of two sources even when their directions are similar.

R5-lite suggests a practical training response. Mixing opponent classes reduces singleton relative leverage, and the gain saturates near equal exposure to pair and singleton opponents. The result motivates opponent-distribution design as a complement to longer self-play. It does not imply that a full AlphaStar-style league is required for the S7 equilibrium itself [2]. In this benchmark, single-policy self-play converges after approximately 1700 iterations. A population or league becomes relevant when the design objective includes robustness to threat compositions that were not present in the training game.

The pre-training contestability gate provides a cheap way to decide whether a proposed physical regime can support such a study. It tests whether the best radar response spans a useful probability interval before expensive optimization begins. In S7, the gate predicted that cross-fire would reduce selected sectors while keeping the game nontrivial; the trained behavior and co-located control followed that direction. This gate is a design diagnostic, not a substitute for training or a proof of equilibrium.

VII. Limitations

First, the physical model uses a single five-sector azimuth grid, two services, fixed site bearings, and a simplified broadband-jammer link budget. The count–geometry decomposition therefore applies to the tested geometry family; it is not a universal scaling law for deployed arrays. Second, the study is simulation-only. Mission drop is a model-level operational endpoint and does not establish fielded detection, tracking, or electronic-support performance. Third, the main S7 equilibrium has three training seeds, but the co-located mechanism control uses one training seed and R5-lite uses one seed per mixture condition. Fourth, R5-lite skips the jammer update on singleton iterations, so its absolute drop values confound opponent exposure with jammer gradient budget. The ratio-level result is the intended readout, while gradient-aligned multi-seed mixing remains necessary for a stronger causal intervention claim.

VIII. Conclusion

We presented a task-level benchmark for adversarial co-adaptation between multifunction radars and cooperative jammers. The benchmark links array-factor physics and finite energy to bearing-specific missions with deadlines, then evaluates both sides with controls that separate natural scheduling loss from adversarial loss.

Across three converged seeds, adding a second jammer reduced learned-radar containment from $63.7\% \pm 1.0\%$ in the valid two-seed S6 baseline to $23.0\% \pm 1.1\%$ at the same S7 team budget, while idle-jammer radar loss remained low. A co-located control showed that two jammers under the separated S7 radar geometry already produce most of the reduction, with cross-fire adding a smaller penalty. Longer continuation training confirmed a stationary high-loss equilibrium rather than a non-transitive training cycle. Finally, a lightweight opponent-class mixture reduced singleton relative leverage with diminishing returns near 50% exposure.

The practical implication is direct: adversarial radar policies should be evaluated against attacker-count and opponent-composition changes, not only against the opponent used during self-play. The next rigorous extension is a gradient-aligned, multi-seed mixture study and a broader geometry family, followed by hardware-in-the-loop validation.

Reproducibility

The repository contains the S6/S7 environments, MAPPO training drivers, fixed train and validation manifests, test gates, final evaluation JSON files, convergence logs, and the data handoff document. The main S7 results use seeds 20260801–20260803 and 3000 iterations per seed; the mechanism control uses seed 20260811; R5-lite uses seeds 20260821–20260823. The report and raw result paths are listed in `DATA_GUIDE.md`. The paper codebase freezes the validated paper baseline with the tag `s7-paper-baseline-20260826`; performance-only optimizations are kept on a separate branch.

References

[1] J. Heinrich and D. Silver, "Deep reinforcement learning from self-play in imperfect-information games," *arXiv preprint arXiv:1603.01121*, 2016. [Online]. Available: https://arxiv.org/abs/1603.01121

[2] O. Vinyals, I. Babuschkin, W. M. Czarnecki *et al.*, "Grandmaster level in starcraft ii using multi-agent reinforcement learning," *Nature*, vol. 575, no. 7782, pp. 350–354, 2019. [Online]. Available: https://doi.org/10.1038/s41586-019-1724-z

[3] T. Wang, H. Dong, V. Lesser, and C. Zhang, "Roma: Multi-agent reinforcement learning with emergent roles," in *Proceedings of the 37th International Conference on Machine Learning*, vol. 119, 2020, pp. 9876–9886. [Online]. Available: https://arxiv.org/abs/2003.08039

[4] B. Wu, Q. Fu, J. Liang, P. Qu, X. Li, L. Wang, W. Liu, W. Yang, and Y. Liu, "Hierarchical macro strategy model for moba game ai," *arXiv preprint arXiv:1812.07887*, 2018. [Online]. Available: https://arxiv.org/abs/1812.07887

[5] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of ppo in cooperative, multi-agent games," in *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, vol. 35, 2022, pp. 24 611–24 624. [Online]. Available: https://arxiv.org/abs/2103.01955

[6] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017. [Online]. Available: https://arxiv.org/abs/1707.06347